Name: Vanshika Wadhwani
Class: D20A
Roll no: 35

Experiment: 7

Aim:

To understand the Truffle framework, establish a local Ethereum blockchain using Ganache, and compile, deploy, and interact with a Solidity smart contract on the local blockchain.

Theory:

Truffle is a development framework used for building, compiling, testing, and deploying Ethereum-based smart contracts. It provides a structured project environment containing directories for smart contracts, deployment migrations, tests, and configuration.

Ganache is a local blockchain environment used for Ethereum development and testing. It provides local blockchain accounts with test Ether and allows transactions and smart contracts to be executed without connecting to a public Ethereum network.

In this experiment, Truffle is used to create and configure a blockchain project. A Solidity smart contract named SimpleStorage is created to store and retrieve an integer value. The contract is compiled using Truffle and deployed to the local Ganache blockchain through a migration script.

The overall process is:

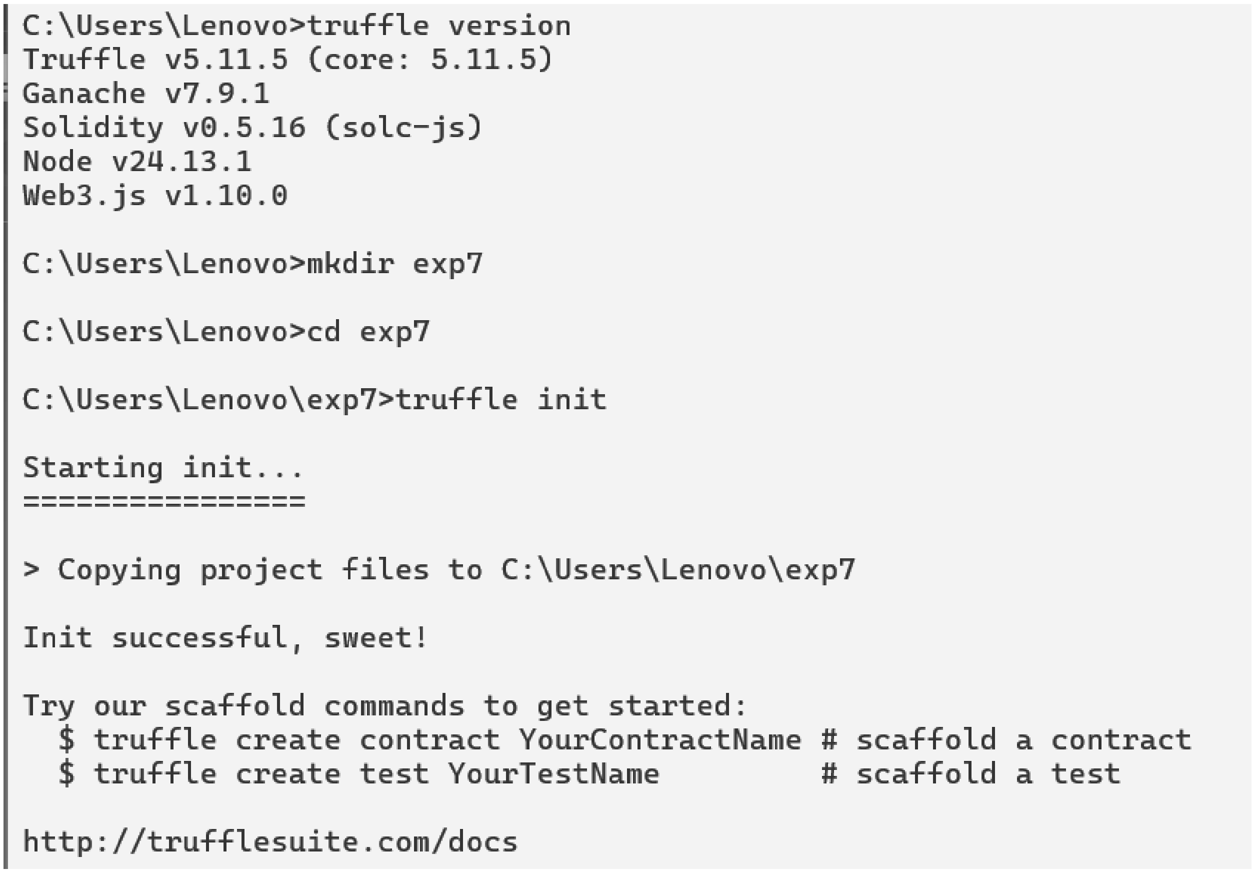
Truffle Project
↓
Smart Contract Creation
↓
Contract Compilation
↓
Start Ganache Local Blockchain
↓
Truffle Network Configuration
↓
Contract Deployment
↓
Contract Interaction

The smart contract contains two main functions:

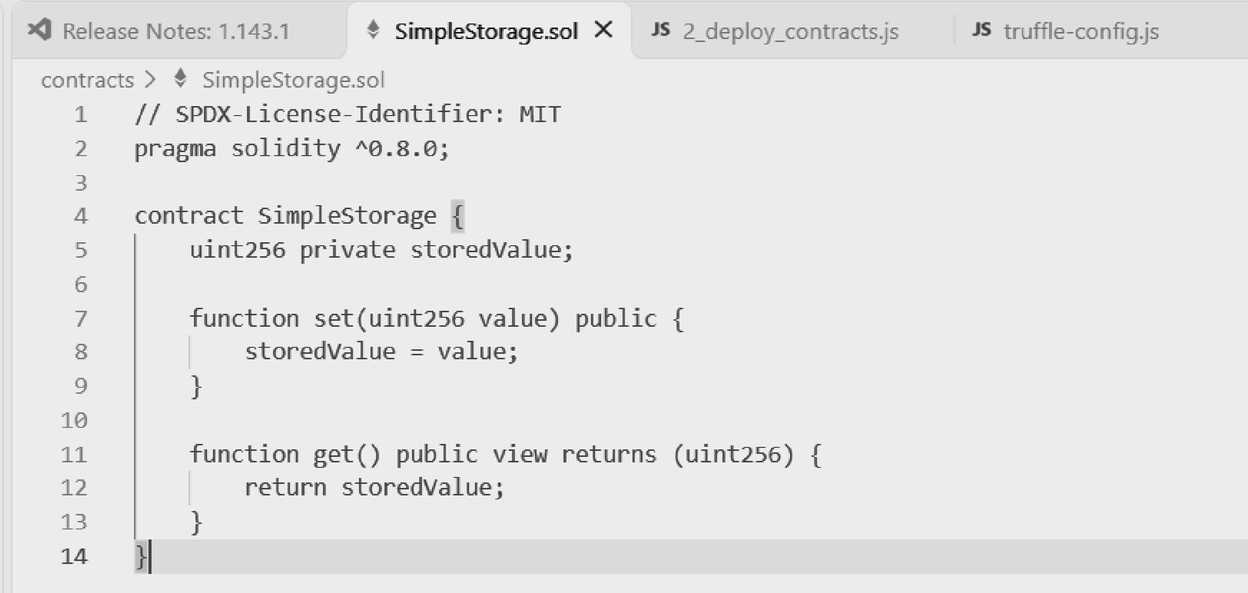
- set(uint256 value) – stores a value in the blockchain.
- get() – retrieves the stored value.

Once deployed, the contract receives a unique contract address on the local blockchain. Transactions generated while deploying and interacting with the contract are recorded as blockchain transactions.

Output:

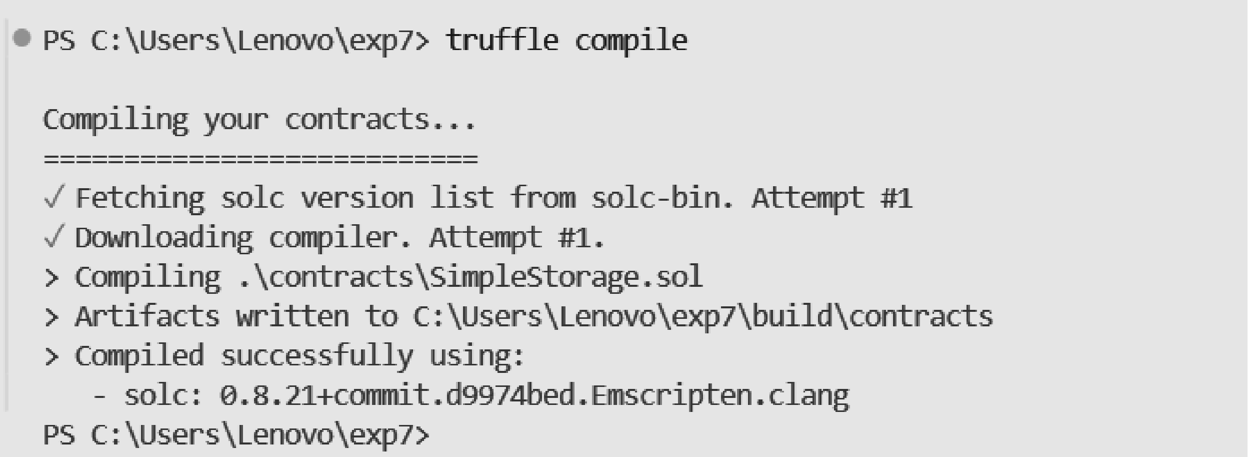


```
C:\Users\Lenovo>truffle version
Truffle v5.11.5 (core: 5.11.5)
Ganache v7.9.1
Solidity v0.5.16 (solc-js)
Node v24.13.1
Web3.js v1.10.0

C:\Users\Lenovo>mkdir exp7

C:\Users\Lenovo>cd exp7

C:\Users\Lenovo\exp7>truffle init

Starting init...
================

> Copying project files to C:\Users\Lenovo\exp7

Init successful, sweet!

Try our scaffold commands to get started:
  $ truffle create contract YourContractName # scaffold a contract
  $ truffle create test YourTestName         # scaffold a test

http://trufflesuite.com/docs
```

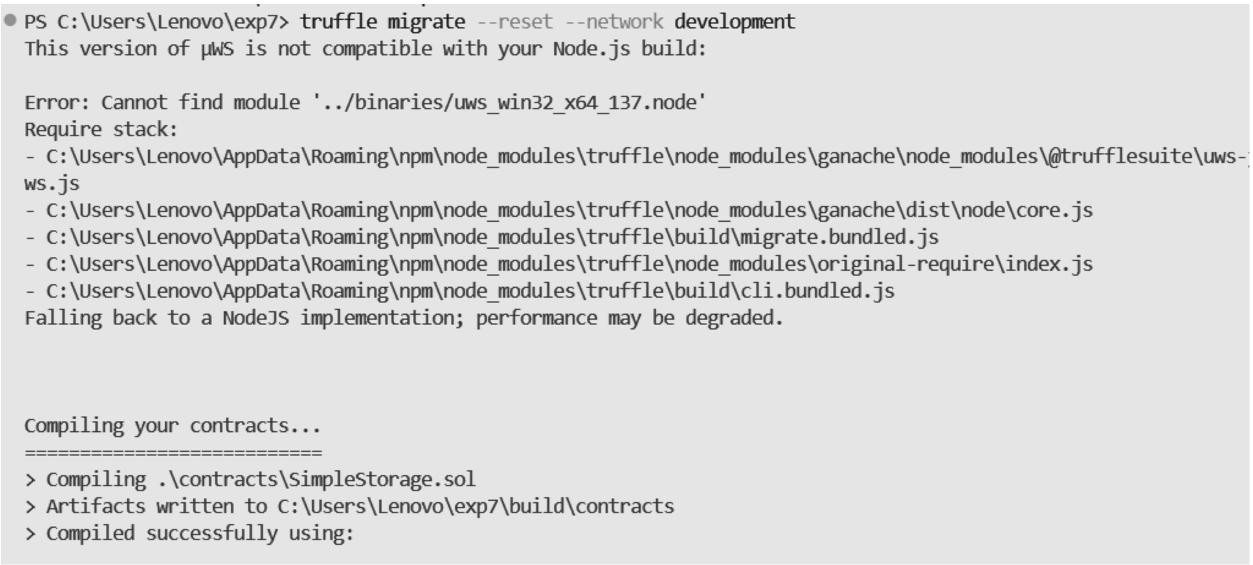


Release Notes: 1.143.1
SimpleStorage.sol
2_deploy_contracts.js
truffle-config.js
contracts > SimpleStorage.sol

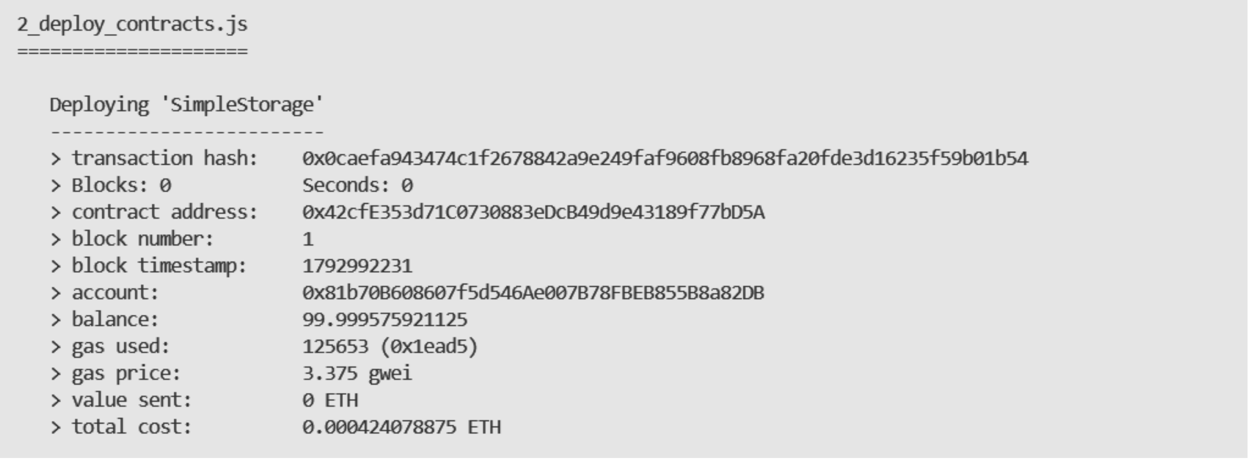
```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.0;

contract SimpleStorage {
    uint256 private storedValue;

    function set(uint256 value) public {
        storedValue = value;
    }

    function get() public view returns (uint256) {
        return storedValue;
    }
}
```



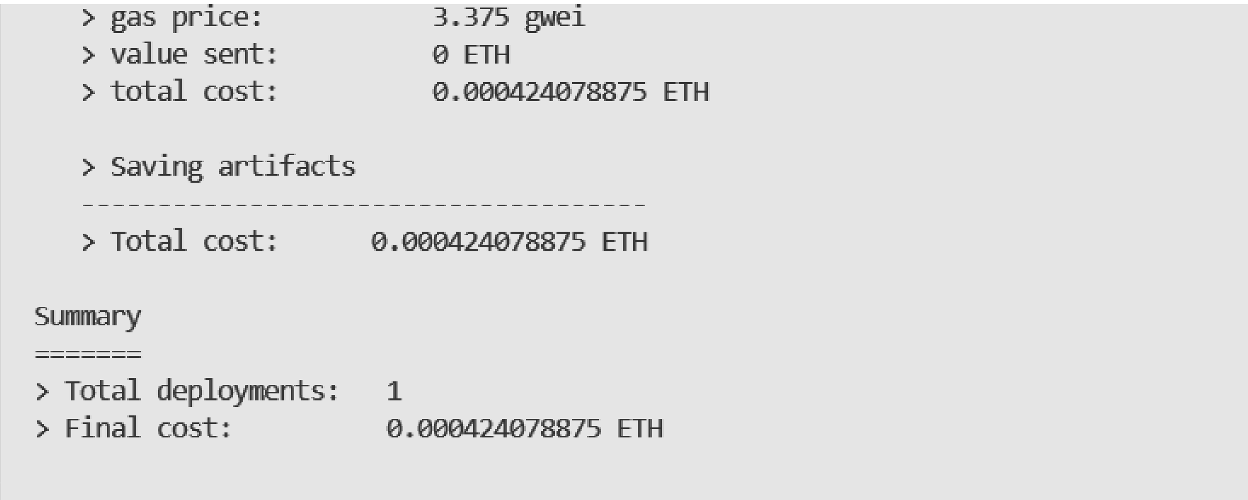
```
● PS C:\Users\Lenovo\exp7> truffle compile

  Compiling your contracts...
  ===========================
  √ Fetching solc version list from solc-bin. Attempt #1
  √ Downloading compiler. Attempt #1.
  > Compiling .\contracts\SimpleStorage.sol
  > Artifacts written to C:\Users\Lenovo\exp7\build\contracts
  > Compiled successfully using:
     - solc: 0.8.21+commit.d9974bed.Emscripten.clang
  PS C:\Users\Lenovo\exp7>
```



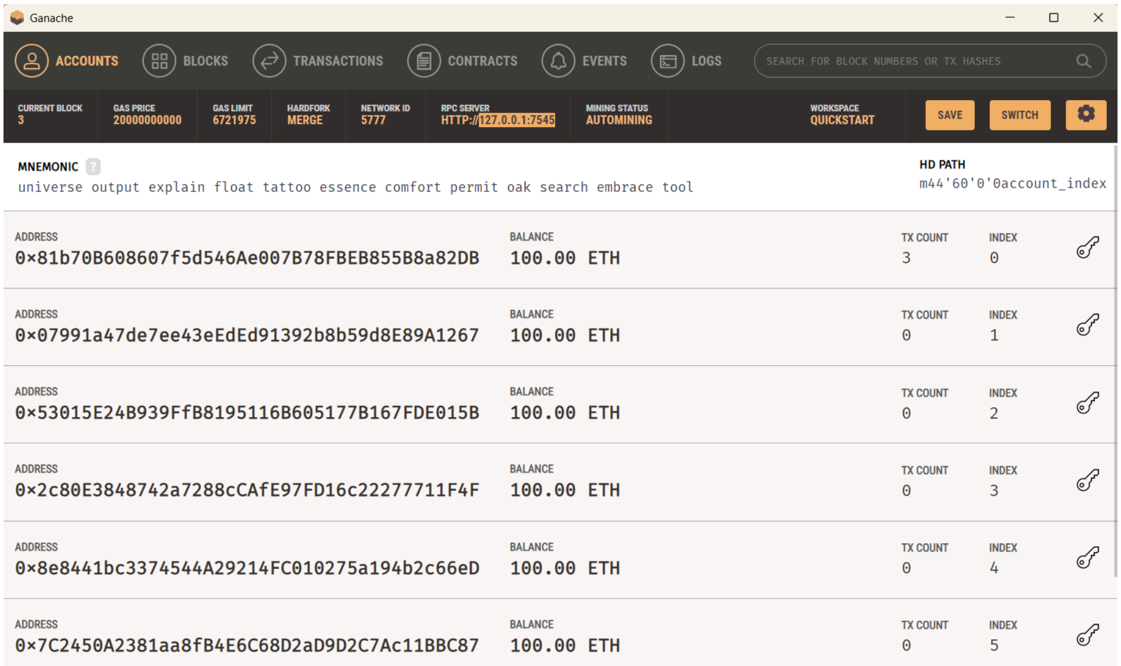
```
● PS C:\Users\Lenovo\exp7> truffle migrate --reset --network development
  This version of μWS is not compatible with your Node.js build:

  Error: Cannot find module '../binaries/uws_win32_x64_137.node'
  Require stack:
  - C:\Users\Lenovo\AppData\Roaming\npm\node_modules\truffle\node_modules\ganache\node_modules\@trufflesuite\uws-
  ws.js
  - C:\Users\Lenovo\AppData\Roaming\npm\node_modules\truffle\node_modules\ganache\dist\node\core.js
  - C:\Users\Lenovo\AppData\Roaming\npm\node_modules\truffle\build\migrate.bundled.js
  - C:\Users\Lenovo\AppData\Roaming\npm\node_modules\truffle\node_modules\original-require\index.js
  - C:\Users\Lenovo\AppData\Roaming\npm\node_modules\truffle\build\cli.bundled.js
  Falling back to a NodeJS implementation; performance may be degraded.



  Compiling your contracts...
  ===========================
  > Compiling .\contracts\SimpleStorage.sol
  > Artifacts written to C:\Users\Lenovo\exp7\build\contracts
  > Compiled successfully using:
```



```
2_deploy_contracts.js
=====================

   Deploying 'SimpleStorage'
   -------------------------
   > transaction hash:    0x0caefa943474c1f2678842a9e249faf9608fb8968fa20fde3d16235f59b01b54
   > Blocks: 0            Seconds: 0
   > contract address:    0x42cfE353d71C0730883eDcB49d9e43189f77bD5A
   > block number:        1
   > block timestamp:     1792992231
   > account:             0x81b70B608607f5d546Ae007B78FBEB855B8a82DB
   > balance:             99.999575921125
   > gas used:            125653 (0x1ead5)
   > gas price:           3.375 gwei
   > value sent:          0 ETH
   > total cost:          0.000424078875 ETH
```



```
    > gas price:          3.375 gwei
    > value sent:         0 ETH
    > total cost:         0.000424078875 ETH

    > Saving artifacts
    -------------------------------------
    > Total cost:      0.000424078875 ETH

Summary
=======
> Total deployments:   1
> Final cost:          0.000424078875 ETH
```

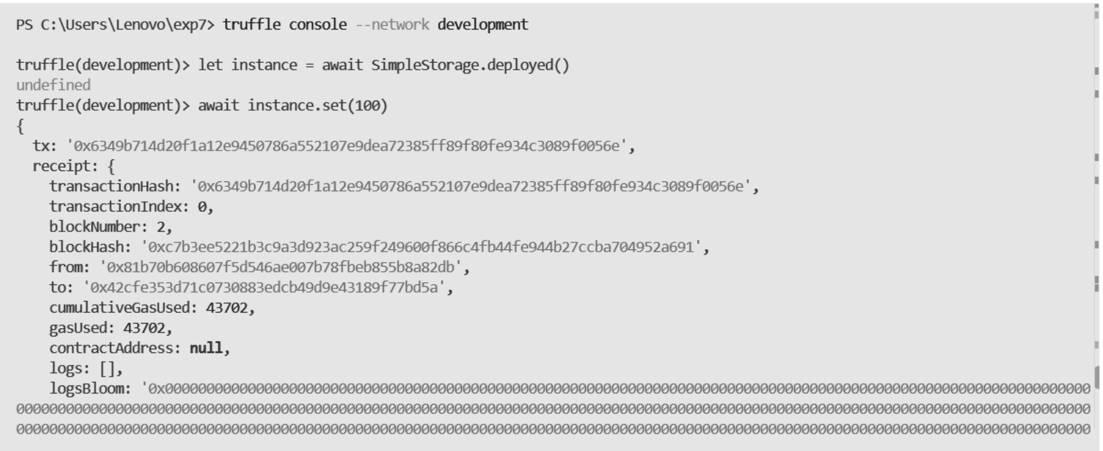


CURRENT BLOCK 3
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART

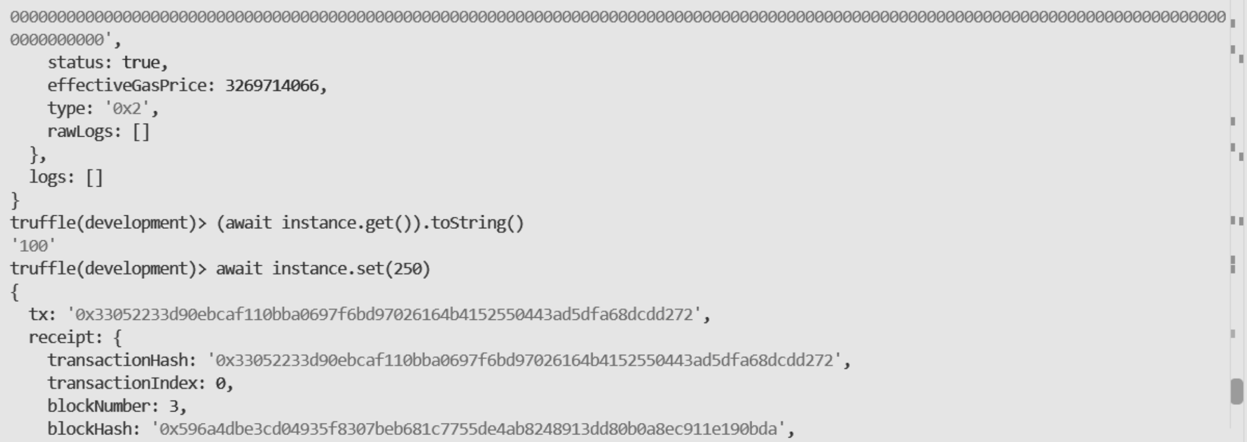
MNEMONIC
universe output explain float tattoo essence comfort permit oak search embrace tool

HD PATH
m44'60'0'0account_index

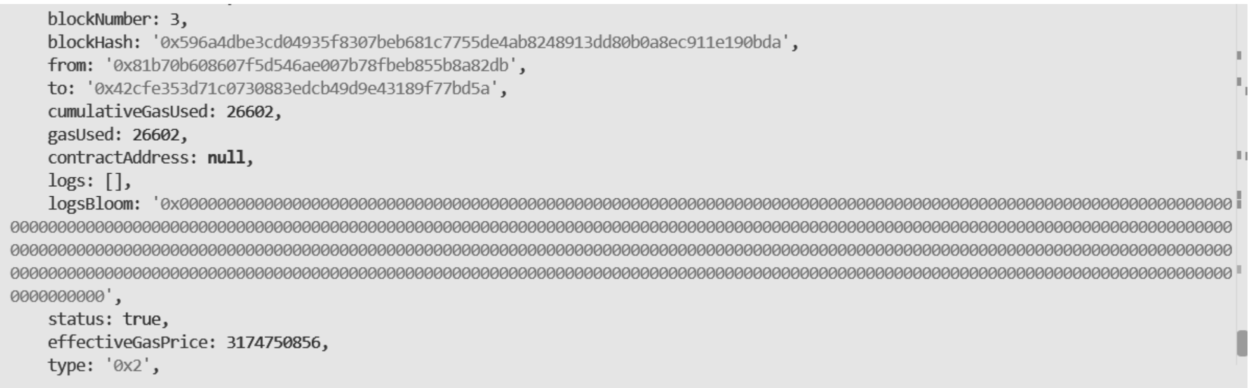
ADDRESS	BALANCE	TX COUNT	INDEX
0×81b70B608607f5d546Ae007B78FBEB855B8a82DB	100.00 ETH	3	0
0×07991a47de7ee43eEdEd91392b8b59d8E89A1267	100.00 ETH	0	1
0×53015E24B939FfB8195116B605177B167FDE015B	100.00 ETH	0	2
0×2c80E3848742a7288cCAfE97FD16c22277711F4F	100.00 ETH	0	3
0×8e8441bc3374544A29214FC010275a194b2c66eD	100.00 ETH	0	4
0×7C2450A2381aa8fB4E6C68D2aD9D2C7Ac11BBC87	100.00 ETH	0	5



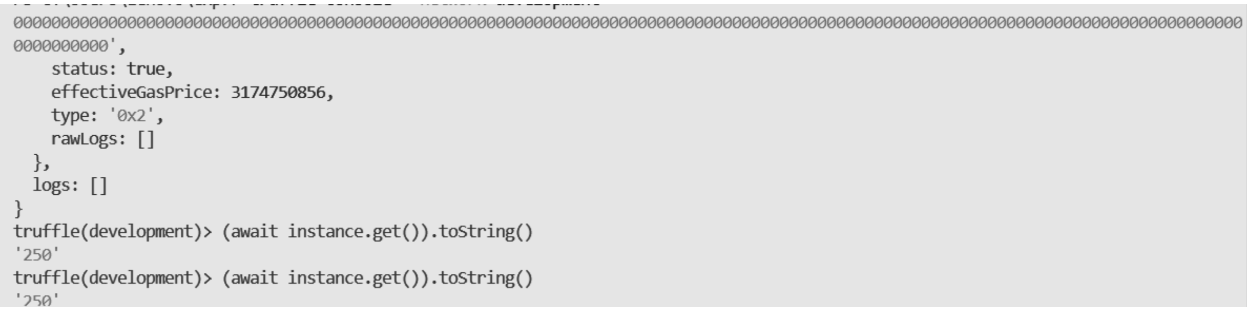
```
PS C:\Users\Lenovo\exp7> truffle console --network development

truffle(development)> let instance = await SimpleStorage.deployed()
undefined
truffle(development)> await instance.set(100)
{
  tx: '0x6349b714d20f1a12e9450786a552107e9dea72385ff89f80fe934c3089f0056e',
  receipt: {
    transactionHash: '0x6349b714d20f1a12e9450786a552107e9dea72385ff89f80fe934c3089f0056e',
    transactionIndex: 0,
    blockNumber: 2,
    blockHash: '0xc7b3ee5221b3c9a3d923ac259f249600f866c4fb44fe944b27ccba704952a691',
    from: '0x81b70b608607f5d546ae007b78fbeb855b8a82db',
    to: '0x42cfe353d71c0730883edcb49d9e43189f77bd5a',
    cumulativeGasUsed: 43702,
    gasUsed: 43702,
    contractAddress: null,
    logs: [],
    logsBloom: '0x00000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
00000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
```



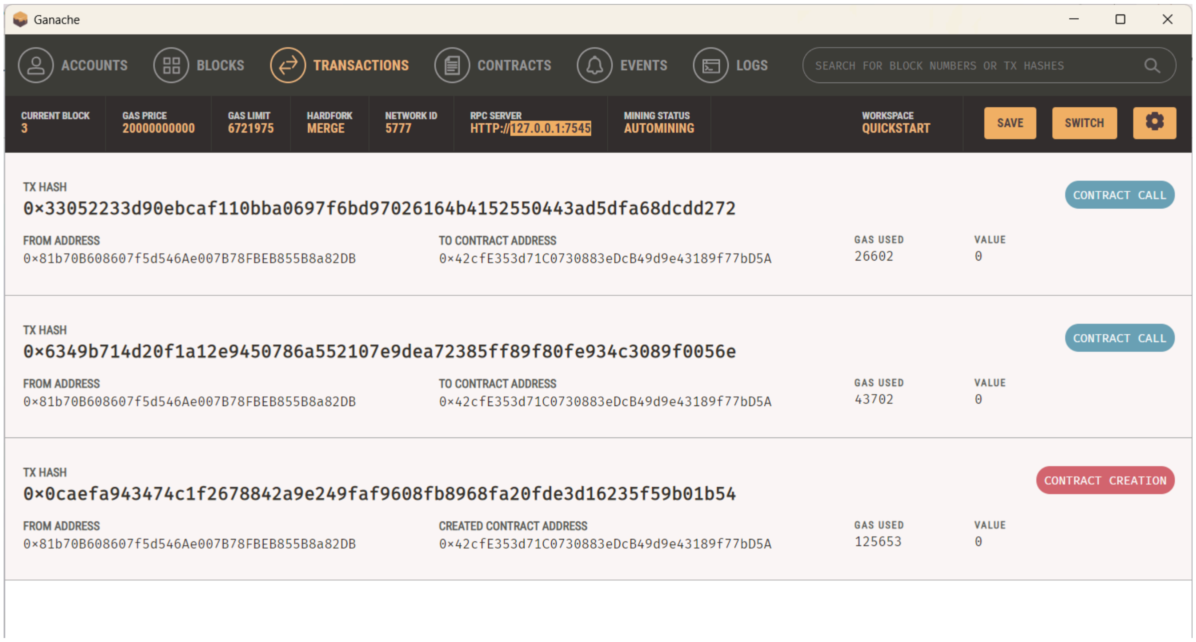
```
000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000',
    status: true,
    effectiveGasPrice: 3269714066,
    type: '0x2',
    rawLogs: []
  },
  logs: []
}
truffle(development)> (await instance.get()).toString()
'100'
truffle(development)> await instance.set(250)
{
  tx: '0x33052233d90ebcaf110bba0697f6bd97026164b4152550443ad5dfa68dcdd272',
  receipt: {
    transactionHash: '0x33052233d90ebcaf110bba0697f6bd97026164b4152550443ad5dfa68dcdd272',
    transactionIndex: 0,
    blockNumber: 3,
    blockHash: '0x596a4dbe3cd04935f8307beb681c7755de4ab8248913dd80b0a8ec911e190bda',
```



```
    blockNumber: 3,
    blockHash: '0x596a4dbe3cd04935f8307beb681c7755de4ab8248913dd80b0a8ec911e190bda',
    from: '0x81b70b608607f5d546ae007b78fbeb855b8a82db',
    to: '0x42cfe353d71c0730883edcb49d9e43189f77bd5a',
    cumulativeGasUsed: 26602,
    gasUsed: 26602,
    contractAddress: null,
    logs: [],
    logsBloom: '0x0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000',
    status: true,
    effectiveGasPrice: 3174750856,
    type: '0x2',
```



```
0000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000000
0000000000',
    status: true,
    effectiveGasPrice: 3174750856,
    type: '0x2',
    rawLogs: []
  },
  logs: []
}
truffle(development)> (await instance.get()).toString()
'250'
truffle(development)> (await instance.get()).toString()
'250'
```


Ganache
ACCOUNTS
BLOCKS
TRANSACTIONS
CONTRACTS
EVENTS
LOGS
SEARCH FOR BLOCK NUMBERS OR TX HASHES
SAVE
SWITCH
CONTRACT CALL
CONTRACT CALL
CONTRACT CREATION

CURRENT BLOCK 3
GAS PRICE 20000000000
GAS LIMIT 6721975
HARDFORK MERGE
NETWORK ID 5777
RPC SERVER HTTP://127.0.0.1:7545
MINING STATUS AUTOMINING
WORKSPACE QUICKSTART

TX HASH 0×33052233d90ebcaf110bba0697f6bd97026164b4152550443ad5dfa68dcdd272
FROM ADDRESS 0×81b70B608607f5d546Ae007B78FBEB855B8a82DB
TO CONTRACT ADDRESS 0×42cfE353d71C0730883eDcB49d9e43189f77bD5A
GAS USED 26602
VALUE 0

TX HASH 0×6349b714d20f1a12e9450786a552107e9dea72385ff89f80fe934c3089f0056e
FROM ADDRESS 0×81b70B608607f5d546Ae007B78FBEB855B8a82DB
TO CONTRACT ADDRESS 0×42cfE353d71C0730883eDcB49d9e43189f77bD5A
GAS USED 43702
VALUE 0

TX HASH 0×0caefa943474c1f2678842a9e249faf9608fb8968fa20fde3d16235f59b01b54
FROM ADDRESS 0×81b70B608607f5d546Ae007B78FBEB855B8a82DB
CREATED CONTRACT ADDRESS 0×42cfE353d71C0730883eDcB49d9e43189f77bD5A
GAS USED 125653
VALUE 0

Conclusion:

Thus, the Truffle framework was successfully used to create and configure a blockchain project. A local Ethereum blockchain was established using Ganache, and the SimpleStorage smart contract was successfully compiled, deployed, and tested. The stored value was successfully retrieved from the deployed contract, demonstrating the basic process of smart contract development and deployment on a local blockchain.